Exam technique for the VWO mathematics central exam

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A strong mathematician can still lose 15–20% of the points on a Dutch central exam by not writing down what the marker needs. This page explains how points are awarded and how to collect all of them.

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1. How marks are awarded

Each question has a fixed number of points (usually 3–6). The *correctievoorschrift* (marking scheme) splits them into bullets, one point each, for example:

- inzicht dat $f'(x) = 0$ moet worden opgelost — **1**
- $f'(x)$ correct — **1**
- beschrijven hoe de vergelijking met de GR wordt opgelost — **1**
- het antwoord — **1**

Consequences:

- **The final answer is often worth only one point.** Most points are for visible steps.
- **"Inzicht" (insight) points** are given for writing down the plan: "the tangent is horizontal, so $f'(x) = 0$".
- Each paper is marked by your own teacher and by a second marker from another school (*tweede corrector*). Neither can give points for steps that are only in your head.
- The general rules at the start of each *correctievoorschrift* apply to every question: no working means no points (even with a correct answer); wrong rounding costs a point; missing calculator explanation costs a point.

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2. Command words

The command word (*opdrachtwoord*) tells you what kind of answer is required. These are the ones in VWO mathematics exams.

Dutch	Meaning	What you must write	Calculator?
Bereken	calculate	The method, the equation or expression, what you entered in the GR, and the result rounded as asked	Allowed
Bereken exact / **exact**	calculate exactly	An algebraic derivation; the answer contains fractions, roots, π, e, ln, etc., not decimals	Only to check; a GR result earns no points
Toon aan	show that	A complete derivation of the given result. Every step that leads to it must be on paper	Allowed, unless "exact" or "algebraïsch" is added; see below

Dutch	Meaning	What you must write	Calculator?
Bewijs	prove	A logical argument with reasons (geometry: *gegeven / te bewijzen / bewijs*, a reason for each step)	Not useful
Benader	approximate	A numerical method (calculator, Euler's method, a table), with the method described	Allowed
Beredeneer / Licht toe / Verklaar	reason / explain	A short argument in words (and formulas), without necessarily computing everything	Usually not needed
Onderzoek	investigate	Find out whether something holds, and give a conclusion with justification	Allowed unless "exact"
Stel een formule op / Druk uit in	set up a formula / express in	A formula, derived and simplified as asked, using the given variables	No
Geef / Schrijf	give / write	The answer; little or no working needed (read the question carefully)	—
Teken / Schets	draw / sketch	A clear figure, with the relevant points labelled	—
Algebraïsch	algebraically	Like "exact": solve by algebra, not by GR	Only to check

Special rules for "toon aan"

- **The given answer must not be used.** Starting from what you are asked to show (and "checking" it) is circular and earns little.
- **Approximate values:** if asked to show that something is "ongeveer 2,3" (about 2.3), give your result **in more decimals** than given (e.g. 2,31…). Otherwise the marker cannot see that you computed it and did not copy it.
- **Equations:** if asked to show that a solution is $x = 3$, it is not enough to substitute $x = 3$ unless the question asks "toon aan dat $x = 3$ een oplossing is". Solve the equation.

"Exact" questions

- Use the GR only to check (and to see what the answer should be).
- Simplifying is good but rarely required: $\frac{2}{\sqrt{2}}$ is accepted, 1.414 is not.

- Exact trig values: know sin and cos of multiples of $\frac{\pi}{6}$ and $\frac{\pi}{4}$ by heart.
- An exact answer may still need a final decimal approximation if the question asks for one ("bereken exact en rond af op…" is rare, but read it).

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3. Writing down calculator work

For every *bereken* question solved with the GR, write:

1. **What you are solving**, in mathematical form: "$f(x) = 3$" or "$f'(x) = 0$" or "$\int_0^4 g(x)\,dx$".
2. **What you entered**: "$Y_1 = xe^{-0.5x}$, $Y_2 = 0.5$, intersect" or "fMin on $[0, 5]$" or "normalcdf(9.9, 10.1, 10, 0.05)". Use mathematical notation where you can; exact key names are not required.
3. **The result**, with enough decimals, then rounded as asked.

Missing step 2 typically costs one point. Writing only "GR: $x = 2.35$" is not enough.

Common calculator traps.

- **Degrees vs radians.** Trigonometric functions in radians unless the question uses degrees. Check the mode at the start of the exam.
- **Window.** An intersection or zero outside the window is missed. Read off the domain from the question and set the window accordingly; look at a table to find all solutions.
- **Numerical derivative** (nDeriv, $\frac{d}{dx}$) is fine for *bereken*, never for *exact* or *toon aan* where the derivative itself is asked.
- **Storing values.** Store intermediate results in variables (A, B, …) instead of retyping rounded values.

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4. Rounding

- **Round only at the end.** Use unrounded intermediate values (stored in the calculator).
- **Round to what is asked**: "op één decimaal" (one decimal), "in hele graden" (whole degrees), "op duizendsten" (three decimals). If nothing is asked, use the context: money in cents, people as whole numbers, probabilities in three or four decimals.
- **Context matters for direction.** "How many whole minutes until the level is *below* 40?" means rounding *up*, even if the decimal part is small.
- **Dutch decimal comma.** Exams use $2{,}35$. Either notation is accepted in your own work, but read the paper carefully: "1,5" means one and a half, and "1.500" can mean one thousand five hundred.

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5. Showing work in common question types

Question type	Minimum working
Extreme value	$f'(x) = 0$ stated; $f'(x)$ written out; solution; why it is a max/min (sign change, graph, or GR); the requested value (often the y-value, not x)
Tangent line	$f'(a)$ as slope; point $(a, f(a))$; equation $y = mx + b$ with b computed

Question type	Minimum working
Area / volume	The integral written with limits; for exact: the antiderivative; for bereken: GR input
Trig equation	Isolate sin/cos; **all** solutions in the domain (use the unit circle, $+k \cdot 2\pi$, then select)
Angle between lines/vectors	Direction or normal vectors; the inner product formula filled in; the acute angle if lines
Parameter questions	Equation in the parameter; solve; check that each solution actually satisfies the condition
Probability	Name the distribution and parameters ("X binomial, $n = 50$, $p = 0.046$"); write the probability as $P(X \geq 3) = 1 - P(X \leq 2)$; the GR command
Proof	*Gegeven*, *te bewijzen*, numbered steps each with a reason

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6. The most common ways to lose points

1. **No working for a correct answer.** Especially in *bereken* questions solved entirely on the GR.
2. **Approximating in an *exact* question**, e.g. using 1.414 for $\sqrt{2}$ in the middle of a derivation.
3. **Missing solutions** of trig equations or of $|x|$-type equations; or not selecting solutions in the given domain.
4. **Answering a different question.** Asked for the y-coordinate, gave the x-value. Asked for "the time", gave the volume. Underline what is asked before you start.
5. **Rounding too early**, so the last decimal is wrong.
6. **Wrong rounding direction** in context ("at least", "less than").
7. **Circular *toon aan*** that starts from the given result.
8. **Radians/degrees mix-up.**
9. **Forgetting units** in context questions (minutes, cm², euros).
10. **Spending 25 minutes on one 4-point question** and not reaching the last two pages. Use the time budget (about 2.4 minutes per point).
11. **Not checking the domain** (e.g. ln of a negative number, a zero outside $[0, 2\pi]$).
12. **Messy layout.** If the marker cannot follow it, points are lost. Write one step per line, and cross out cleanly (a single line) instead of overwriting.

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7. On the day

- Check the calculator (exam mode, batteries, radians) before the exam starts.
- Read the whole paper first; start with a part you like to build confidence.
- Keep track of time at each part boundary: after 60 minutes you should be about one third of the way through the points.
- Leave the last 10 minutes to check: every question answered? Every answer rounded as asked? Units?
- A partly answered question still earns points: never leave a question blank. Write the plan ("$f'(x) = 0$ solve…") even if you cannot finish it.